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11 September 1987
SUPERSEDING
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25 May 1978

# MILITARY SPECIFICATION

## LATHES, TURRET, HORIZONTAL, UNIVERSAL, RAM TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers electrical motor-driven, ram type, horizontal, universal turret lathes for manufacturing duplicate parts to precision tolerances from bars fed through the spindle and from castings and forgings held in a chuck.

1.2 Classification. The turret lathes covered by this specification are of the following sizes. The size furnished shall be as specified (see 6.2.1).

| Size | Bar capacity |
|---|---|
| 2 | 2-inch bar diameter, round |
| 3 | 3-inch bar diameter, round |
| 5 | 5-inch bar diameter, round |

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposed (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

2.1.2 Other Government documents and publications. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified. the issues shall be those in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD-adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the non-Government documents which are current on the date of solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA MG 1 - Motors and Generators.

ANSI/NFPA 70 - National Electrical Code.



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ANSI/NFPA 79 - Electrical Standards for Industrial Machinery.

ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues.

ANSI B5.8 - Chucks and Chuck Jaws.

ANSI B5.9 - Spindle Noses for Tool Room Lathes, Engine Lathes, Turret Lathes, and Automatic Lathes (ISO 702).

ANSI B5.16 - Accuracy of Engine and Toolroom Lathes.

ANSI B11.6 - Safety Requirements for the Construction, Care, and Use of Lathes.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept. 1430 Broadway, New York, NY 10018-3363.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

AMERICAN GEAR MANUFACTURERS, ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the International Organization for Standardization, c/o American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018-3363.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)



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2.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The lathe shall be new and one of the manufacturer's current models capable of operations in accordance with the requirements herein. The lathe shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. Use of a spacer between the headstock and the bed to achieve the capacity requirements in this specification shall not be acceptable.

3.2.1 Measurement systems. Unless otherwise specified (see 6.2.1), either the U. S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the lathe. In this specification, all measurements, dimensions, sizes, and capacities are given in US units. The measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified (see 6.2.1), either the US or SI units shall be used to graduate measuring and indicating devices. Regardless of the measurement system used, all measuring and indicating devices on the lathe shall be graduated in the same system. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position. When specified (see 6.2.1), indicating dials shall be graduated to read measurements in both US and SI Units.

3.2.1.2 Digital readout (DRO) system. When specified (see 6.2.1), a DRO system shall be provided on the X and Z axes of travel. The DRO system shall consist of a position feedback device, an electronic display for each axis, and all related components, mounting hardware, and electrical equipment necessary to assure complete operation of the system. The system shall provide a readily visible digital display over the full working range of each axis on which installed. The system resolution shall be 0.0001 inch. The system accuracy shall be equal to the system resolution, verified upon installation and calibration of each system on the machine. The repeatability shall be 0.0001 inch, measured over the full travel on each axis. Special requirements for the DRO system components shall be specified (see 6.2.1). The DRO shall indicate axis



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location in both the US and SI units. The DRO system shall be provided with electronic and manual calibration to the machine on which it is installed to eliminate the effects of distortion error.

3.2.1.2.1 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The connecting cable from the position feedback device shall be electronically shielded to protect against electronic interference and shall be armored to protect against contamination and inadvertent damage. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. Installation of the position feedback device shall not limit machine travel on any axis nor eliminate manual machine stops. Provisions shall be made to prevent damage to the feedback device in the event of overtravel. The feedback unit and related components shall be protected from contamination of oil, dust, chips, coolants, and airborne particles. Position feedback devices shall be easily and quickly interchangeable between like machines in any multiple machine order.

3.2.1.2.2 Electronic display console. The electronic display console for the DRO system shall contain a universal digital display with individual illuminated digits not less than 0.56 inch high with an effective digit height of not less than 0.43 inch. Digits shall be high efficiency displays for high light ambient conditions with wide viewing angle and shall be readable at a distance of not less than 16 feet. The console shall display at least six digits simultaneously. The display console shall also provide keyboard entry, memory, absolute/incremental counting, data preset, power failure indicator, inch/metric conversion, and absolute zero recall. The electronic display console shall be installed so as to be readily accessible and easily read from the operator's work position. The console shall be mounted to withstand shock and vibration generated by the machine. The connecting cable shall be armored and shall be installed so as not to interfere with the normal operations of the machine. The cable shall be securely fastened to the structure of the machine in accordance with the requirements of the contract.

3.2.2 Reclaimed materials. The lathe may contain reclaimed materials provided such materials will not jeopardize its intended use, performance, or design life. The materials shall have been reclaimed in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the lathe.

3.2.3 Energy efficiency. The lathe and its applicable components that directly consume energy in normal operation shall be designed and constructed for the maximum degree of energy efficiency as governed by the developments available within the industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the lathe that present safety



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hazards. The safety devices shall not interfere with the operation of the lathe. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All lathe parts, components, mechanisms, and assemblies furnished on the lathe, whether or not specifically required herein, shall comply with all of the requirements of OSHA 29 CFR 1910 that are applicable to the lathe itself. In addition, the machine shall comply with all requirements of ANSI B11.6 that are designated therein as the responsibility of the machine manufacturer. In the event of a conflict between the requirements of the OSHA and ANSI Standards, the requirements of OSHA shall apply. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Mercury restriction. The lathe shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the lathe or in the provided accessories.

3.2.8 Environmental protection. Under the operating, service, transportation, and storage conditions described herein, the lathe shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. Enclosed headstock and carriage mechanisms having gears, clutches, sliding parts, and bearings that require lubrication shall be serviced by an automatic pressure system, flood system, mist system, or any combination of these methods to ensure adequate lubrication to these and all other moving parts. Each system that recirulates lubricating oil shall include a cleanable or replaceable filter. Each lubricant reservoir shall have a convenient means of determining fluid level. All oil holes, grease fittings, and filler caps shall be easily accessible.

3.2.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances. When industry or Government standards are applicable, all replacement parts shall conform to the standards. All replacement parts shall be interchangeable with the original respective parts provided by the original equipment manufacturer (OEM) without modification to the replacement part(s) or the mating components. All parts subject to possible future replacement as spare or repair parts shall be assigned a manufacturer's part number and shall be identified in the repair parts manual. Foreign-sourced replacement parts shall be interchangable with domestically supplied components or be logistically supportable and readily available from a domestic source. Replacement parts are those normally subjected to wear such as gears, bearings, shafts, feed screws, drive belts, pumps, and seals. Electrical replacement parts include motors, relays, switches, magnetics, pushbuttons, light bulbs, fuses, braking mechanisms, and solid-state devices. Tooling and accessories normally supplied as standard items shall also be considered replacement parts.

3.3 Construction. The lathe shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand



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all forces encountered during operation of the lathe to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the lathe shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the lathe and its related attachments and accessories shall conform to FED-STD-H28 and the applicable 'Detailed Standard' section referenced therein.

3.3.7 Dials. All rotating dials or handwheels used to position machine axes or for tool movement shall be graduated in increments not larger than 0.10 inch for longitudinal and 0.001 inch for cross motion. Such graduations shall accurately indicate machine component movement or material removal. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background and be of such size that graduations can be read from the operator's normal position. Feed dials shall have independent zero adjustments and be calibrated in such a manner that the last dial graduation progresses into and in continuous with the first dial graduation as the dial is rotated through the zero position.

3.3.8 Gears. All gears and pinions of spindle and axis drive trains shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the English (US) system or ISO 54 for the metric (SI) system. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened steel. Hardness shall be not less than Rockwell C-32.



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3.3.9 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The lathe shall consist of not less than a base and bed, headstock, cross-slide carriage, cross slide, square turret, cross-slide apron, hexagon turret saddle, sliding ram, hexagon turret, and turret apron.

3.4.1 Bed and base. The lathe shall have a bed and base structure that supports the headstock, tailstock, carriage, and other related equipment. The bed shall be a stress-relieved ferrous casting with sufficient strength and rigidity to maintain accurate alignment of all the components in accordance with ANSI B5.16. The bed shall have V-ways, flat ways, or a combination of V-ways and flat ways to provide a sliding surface for the longitudinal movement of the carriage and assure alignment of the headstock and tailstock. The bed ways shall be cast integrally with the bed structure, or of the replaceable way inserts type, or a combination thereof. The bed ways for the carriage longitudinal movement shall be hardened to Rockwell C-50, if integral, or Rockwell C-55, if inserted, and ground to minimize wear and maintain accuracy of alignment. Means shall be provided to lubricate the bed ways automatically or with a one-shot lubrication method as specified (see 6.2.1). The bed shall have a sufficient number of covered access openings to facilitate inspection, adjustment, and repair of components located within the structure. The base shall be provided with means for securing and leveling the machine on a foundation.

3.4.2 Headstock. The lathe shall have a headstock that supports and aligns the spindle axis parallel with the bed ways. It shall contain all the gearing necessary to provide the required speeds; clutch and quick acting brake mechanism to start and stop the spindle rotation and to allow free rotation for changing speeds; and a power takeoff to the gearbox for feeding and threading. The headstock shall house the gearing for the spindle drive and the power take-off source for the feed mechanism. Speed selections shall be made by manually setting levers or dials on the headstock. Each lathe shall have a clutch and brake mechanism to provide smooth starting and braking of the headstock spindle rotation and to allow free rotation for changing spindle speeds. Spindle start and stop shall be by levers or knobs on the headstock and on the apron of the lathe. The headstock shall support the spindle and gear shafts to prevent deflection that would affect accuracy under full operating and cutting loads. All load carrying surfaces of gears, clutch jaws, and surfaces of shafts on which gears and clutches slide, or internal surfaces of sliding parts, shall have a surface hardness of not less than Rockwell C-48. When power to the spindle drive mechanism is transmitted through V-belts, means shall be provided to maintain proper belt tension. The headstock shall be oiltight and shall have automatic lubrication to all internal parts that require lubrication. When specified (see 6.2.1), a temperature sensor shall be provided to monitor the temperature of the headstock lubricant and shut the machine off at a preset temperature limit below the flashpoint of the oil vapor.



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3.4.3 Headstock spindle. The lathe shall have a headstock spindle to provide accurate mounting and rotation of a chuck, faceplate, or collet. The spindle shall be supported by roller or ball-type bearings of a size, type, and precision class suitable for the application. Unless otherwise specified (see 6.2.1), the spindle nose type shall be type 'A1' or 'A2', as normally provided and as specified in table I, and shall conform to the requirements of ANSI B5.9. The spindle nose shall be accurately machined to provide locating surfaces for chucks and collets.

3.4.3.1 Spindle drive. The spindle shall be driven through either an all gear or combination gear and belt drive by an electric motor as specified in table I. The combination drive shall provide nonslip positive belt drive in the high-speed range; the intermediate and low speeds shall be gear driven. Spindle speeds shall be in approximate geometric progression. The spindle shall operate in both directions of rotation.

3.4.3.2 Spindle speed change mechanism. The spindle speed mechanism shall be manual speed change or manual speed change with preselect power shift, as specified (see 6.2.1). The shifting mechanism shall be designed so that speed change cannot occur unintentionally. Except for the variable speed drive, speeds shall be in geometric progression within 10 percent. The manual speed change mechanism shall allow spindle speed changes by positioning gear shift levers or dials located on the front side of the headstock. Speeds shall be selected by positioning the levers as indicated on a conveniently located speed chart or by positioning a direct reading dial. The manual speed change with preselect power shift shall provide spindle speed selection by manually setting a selector control to the desired spindle speed. It shall be possible to preselect the next spindle speed while still cutting, then automatically initiate the shift upon activation of the spindle control. The actual spindle speed in revolutions per minute (RPM) shall be shown by a speed indicating device located convenient to the operator.

3.4.4 Carriage. The carriage shall support the apron, cross slide, and compound rest. The carriage shall traverse longitudinally on the bed ways. Means shall be provided to maintain adjustment between sliding surfaces and securely lock the carriage in position on the bed when cross feeding. The carriage shall operate in both longitudinal directions under power, regardless of the direction of the spindle rotation. Way wipers shall be fitted to the carriage wings to prevent entrance of chips and abrasive materials between the way surfaces. Provisions shall be made for use of a taper attachment and follow rest.

3.4.5 Apron. The lathe shall have an apron on the front side of the carriage which shall contain the feed mechanism, threading mechanism, operating controls, and lubrication system for the apron and carriage components. The operating controls shall include power longitudinal and cross feeds, automatically operated longitudinal feed stops, threading controls, and a spindle start and stop. Lever type operating controls, such as the threading and feed engagement, shall fully engage and disengage with smooth, positive action. The apron shall be of double wall construction, providing inboard and outboard supports for both ends of gear shafts. Gearing in the apron shall provide hand and power longitudinal and cross feeds. An interlock shall be incorporated in the apron to prevent simultaneous engagement of the feed and half-nut. Handwheels


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shall automatically disengage to assure nonrotation while the carriage is under power. The feed rod and leadscrew shall be separate units.

3.4.6 Cross slide. The cross slide shall be fitted to the carriage with adjustable gibs. The cross slide ways or the carriage ways mating with the cross slide shall be hardened to not less than Rockwell C-45. Threads of the cross slide feed screw shall have a surface hardness of not less than Rockwell C-45, be of the acme or square thread form, and shall be protected from chips and dirt. The cross slide feed screw shall be supported by anti-friction thrust bearings and shall have adjustment for backlash. The feed screws shall be fitted with direct reading micrometer dials.

3.4.7 Hexagon turret saddle. The saddle shall be provided with adjustable gibs to compensate for wear and adjust alignment. The saddle shall have machined ways with bearing surfaces of hardened and ground steel strips for mounting of the sliding ram. The steel bearing strips shall be hardened to not less than Rockwell C-60.

3.4.8 Sliding ram. The ram shall have full bearing the entire length of the ram. Adjustable gibs shall be provided to maintain adjustment of the ram. The ram shall be equipped with a rotating indexing stop to disengage the longitudinal power feed. The stop shall be equipped with not less than six adjustable threaded stop rods with means provided for locking each stop in position. A handwheel with a gear and rack combination shall be provided for longitudinal manual feed of the ram.

3.4.9 Square turret. A square, quick-indexing turret shall be provided consisting essentially of an indexing mechanism, flat seated ground tool holder slots, and clamping set screws. The turret shall be mounted with a tapered locating pin and bushings or curvic coupling to assure indexing of the turret to precise positions. Turret indexing and locking shall be accomplished by means of a single manually operated lever. The turret shall have capacity for indexing four tools to cutting position.

3.4.10 Hexagon turret. The turret shall be automatically locked in cutting position on the forward stroke of the ram, and released for indexing on the rearward stroke. A free position shall permit skip indexing and manual rotation. The turret shall be equipped with a retracting locating pin and matching bushings, or curvic coupling, to assure accurate positioning when indexed. The bushings, if used, shall be hardened to not less than Rockwell C-60. All faces of the turret shall be finish machined for mounting straight round shank and flanged bolt-on tool holders.

3.4.11 Turret apron. The apron shall be fastened to the saddle with means to provide positive location and to prevent movement of the apron on the saddle. The apron shall house the feed gear mechanism. Power feed shall be provided to the ram through a gear train powered by a feed shaft from the headstock. A handwheel shall provide means for hand feeding the turret. A manually operated feed change mechanism shall be provided on the apron with a direct reading dial or chart to indicate the available feeds. A cutting overload safety device shall be provided.



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3.4.12 Electrical system. The electrical system shall conform to ANSI/NFPA 79. Unless otherwise specified (see 6.2.1), each lathe shall draw all of its electrical power from a single 230/460-volt, 3-phase, 60-hertz (Hz) circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The lathe shall be initally wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 70 for the disconnect fuse rating. The wiring in each machine shall have individual point-to-point color coded wires in conformance with ANSI/NFPA 79. Diagrams or schematics shall clearly identify the individual, point-to-point color coded wire and wire termination point location in the electrical system.

3.4.12.1 Motors. Motors shall be machine duty cycle and shall have ball roller bearings of the sealed and permanently lubricated type. Alternating-current (AC) motors shall be designed to operate on 60-Hz, unless otherwise specified (see 6.2.1). Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure, and the requirements stated in ANSI/NEMA MG 1 and ANSI/NFPA 70. Motor horsepower ratings shall conform or exceed the requirements of table I.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the lathe shall be not less than specified in table I for the lathe size specified.

3.6 Alignment and performance tolerances. The machine alignment tolerances shall not exceed the tolerances of table II when checked in accordance with 4.7.2.

3.7 Performance. The lathe shall be capable of machining to the required accuracies when tested in accordance with 4.7.3.

3.8 Worklight. Unless otherwise specified (see 6.2.1), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be attached to provide for adjustment to reflect maximum light on the work area. If a built-in light or lighting system is provided, the adjustable light described above shall not be required.

3.9 Coolant system. A coolant system shall be furnished. It shall include a sump or reservoir, a power driven pump, and piping. The coolant reservoir shall incorporate a chip strainer and a drain plug. The coolant system shall include oil-resistant hoses, tubing, or piping to direct the coolant to tools on the carriage and hexagon turret. A coolant nozzle shall be provided and fitted with a valve to permit the operator to direct and control the amount of coolant flow.

3.10 Standard equipment. Unless otherwise specified (see 6.2.1), the following standard equipment shall be supplied with each lathe.

3.10.1 Power-operated bar feed mechanism. The bar feed mechanism shall include a collet chuck, bar feed unit, bar support stands, stock tube, and stock stop. The feed cycle shall be actuated by a single control device to open the collet, feed stock, and close the collet. The bar support



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stands shall have provisions for leveling and securing to the foundation and shall store extra bars. The stock tube shall minimize noise and permit loading from the machine end of the tube. It shall accommodate a 12-foot long bar when loaded in the machine. The collet chuck grip on the bar shall be adjustable. A complete set of serrated-collet pads shall be supplied in 1/16-inch increments, for holding round, square, and hexagon bars within the size capacity of the lathe.

3.10.2 Chuck, four jaw, independent. The chuck shall be a steel body, four-jaw, independent type with three step, hardened and serrated, reversible top jaws. The chuck shall be the size normally furnished for the size lathe specified and shall conform to the specifications of ANSI B5.8. A self-ejecting chuck key shall be furnished with each chuck.

3.10.3 Chuck guard. The chuck guard shall be a metal cover providing protection from a rotating chuck and serving as a chip guard. The guard shall be easily attached and removed and shall provide access to the chuck.

3.10.4 Special tools. Any special tools normally provided by the machine tool builder for operating and maintaining the lathe shall be supplied.

3.11 Optional equipment. The following optional equipment shall be supplied as specified (see 6.2.1).

3.11.1 Universal scroll chuck. The chuck shall be three-jaw universal scroll chuck of the size normally furnished with the lathe. The chuck shall be heavy duty conforming to specifications of ANSI B5.8 and shall have three-step hardened and serrrated, reversible-top jaws. A self-ejecting chuck key shall be furnished with each chuck.

3.11.2 Power chuck wrench. The chuck wrench shall be electric motor driven and mounted on the lathe so as not to interfere with work or fixtures. The chucking torque shall be adjustable for grip on the work. The motor control shall be mounted convenient to the operator's work station and a safety interlock shall prevent spindle rotation while the wrench is engaged.

3.11.3 Taper attachment. The taper attachment shall be for mounting on the carriage cross slide and shall be graduated to indicate taper in degrees and inches per foot. The taper turned in one setting shall be not less than 3 inches per foot by 6 inches long. A backlash eliminator shall be provided for the cross slide screw.

3.11.4 Leader, follower, and two-speed, two-place, gear box, thread chasing attachment for carriage. The threading attachment shall consist essentially of a two-speed, two-place, gear box to change the pitch of the leader providing two leads for one leader, and matched leaders and followers for each lead of thread required to produce a range of threads from 4 or less to not less than 32 threads per inch. The leader shall be constructed to fit a clamp on the drive shaft and shall be positioned as required to suit part threading. The follower shall be mounted on the feed unit and operated manually by a lever to engage the leader with the follower. An automatic knock-off mechanism shall be provided and shall be

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engaged with and actuated by the feed stop at any predetermined point of travel.

3.11.5 Leading-on attachment for ram. The leading-on attachment shall be for accurately advancing a die head or tap held in the ram turret. Leader and follower sets with the required change gears, or a two-speed gear box, shall provide for cutting standard threads within a range of 4 or less to 36 or more threads per inch. The change gears or gear box shall permit producing threads having a lead different than the leader pitch.

3.11.6 Rapid traverse to carriage cross slide (except size 2). Power for the rapid traverse to the carriage cross slide shall be provided by either a hydraulic system or an electric motor. The power unit shall be attached to the back of the cross-slide carriage, and coupled to the cross-slide screw. The rapid traverse shall be controlled manually by a conveniently located spring loaded lever or switch. Upon release, the lever or switch shall return to a neutral position. The rapid-traverse mechanism shall be provided with safety devices to prevent overtravel of the cross slide. Rapid traverse to the cross slide shall be effective in both directions.

3.12 Repair parts. Such repair parts as are specified (see 6.2.1) shall be furnished.

3.13 Marking on plates. All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. All plates shall be corrosion-resistant.

3.13.1 Lubrication plate. Unless otherwise specified (see 6.2.1), a lubrication plate shall be permanently and securely attached to each machine. The plate shall contain the following information:

Points of lubricant application
Servicing interval for each point
Type of lubricant for each pont
Viscosity of each lubricant
Federal or Military specification number of each lubricant
Size and type of lubricant filter

3.13.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each lathe and contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard lathe and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)



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Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.14 Hour meter. Unless otherwise specified (see 6.2.1), the lathe shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the lathe.

3.15 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.16 Workmanship. Workmanship of the lathe shall be commensurate with all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

3.17 Training. Training required by the procuring activity shall be provided as specified.

3.18 Warranty. Warranty requirements shall be as specified in the contract by the procuring activity. The warranty period shall commence upon completion of the installation and final acceptance test of the machine.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.



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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination or any tests shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4.8. Failure of the item to pass the examination, the tests, or the inspections shall be cause for rejection.

4.6 Examination. The lathe shall be examined to determine compliance with all requirements of this specificaton including compliance with all standards cited in section 2.

4.7 Tests.

4.7.1 Operational test. The lathe shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Alignment tolerances test. Each lathe shall be tested to determine that it complies with the alignment tolerance requirements of 3.6.

4.7.3 Performance test. A test part shall be machined to check the performance of the lathe and accessories under normal cutting conditions. The test part configuration shall be the machine tool builder's design to demonstrate turning straight cuts, facing cuts, single-point threading, and automatic feed knock-off for cross and longitudinal turning. The type of cuts shall be applicable to the type equipment to be supplied for bar work, chucking work, or both. Test part material shall be comparable to SAE 1016 or SAE 1113 having not less than 0.13 percent carbon and Brinell hardness not less than 170/202. The bar size shall be the maximum diameter for the size lathe being tested and not less than 12 inches long. Material for holding in a chuck shall be not less than 10 inches in diameter. Spindle speed and feed rates shall be based on the recommended cutting speed for the material used and for carbide tooling. Test cuts shall be checked and shall be within the following tolerances:



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| | |
|---|---|
| Turning cuts | - Round within 0.001 inch TIR, Cylindrical within 0.001 inch per foot |
| Facing cut | - Flat within 0.001 inch concave in 12 inches |
| Single point thread | - Not less than 3/4 - 10 NC-2, effective thread length not less than 1 inch, within limits of go - not go tolerance for class 2 |

4.7.4 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.9 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The turret lathes covered by this specification are intended for use in machine shops where a moderate or high rate of production of duplicate parts is required.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. First article, if required (see 3.1).

d. If machine is to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in specific measurement system (US or SI), state required system; dual dials, if required (see 3.2.1.1).

f. Digital readout system, if required; specify axes of travel required and any special requirements (see 3.2.1.2).

g. Additional safety and health requirements, if required (see 3.2.5).

h. Painting, if different (see 3.3.5).



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i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.9).

j. Automatic one-shot lubrication system, as required (see 3.4.1).

k Headstock temperature sensor, if required (see 3.4.2).

l. Spindle nose type, if different (see 3.4.3).

m. Spindle speed mechanism, as specified (see 3.4.3.2).

n. Electrical system and input power, if different (see 3.4.12).

o. Motors, if different (see 3.4.12.1).

p. Size and capacity, if different (see 3.5).

q. Worklight, if not required (see 3.8).

r. Standard equipment, if different (see 3.10 through 3.10.4).

s. Specify and fully describe optional equipment, if required (see 3.11 through 3.11.6).

t. Specify applicable repair parts, if required (see 3.12).

u. Lubrication plate, if different (see 3.13.1).

v. Nameplate, if different (see 3.13.2).

w. Hour meter, if not required (see 3.14).

x. Inspection conditions, if different (see 4.3)

y. First article inspection, if different (see 4.4).

z. Quality conformance inspection, if different (see 4.5).

aa. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level required and specification number (see 5.1).

6.2.2 Contract data requirements. Required technical data (operating manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.



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6.4 Cross-reference of classification changes. Lathe classification changes from the previous issue of this specification are as follows:

| MIL-L-4533E | MIL-L-4533F |
|---|---|
| Size 1-3/4 | Omitted |
| ---- | Size 2 - New |
| Size 3 | Size 3 |
| Size 4-1/2 | Omitted |
| ---- | Size 5 - New |

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Inspection facilities. The contracting officer should be notified by each contractor in his response to the solicitation of the location where inspection tests are to be performed. It is recommended all inspections be performed within the limits of the continental United States.

6.7 Subject term (keyword) listing.

Electric motor driven
Horizontal
Lathe
Ram type
Turret
Universal



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TABLE I. Sizes and capacities.

| Machine characteristics | Size 2 | Size 3 | Size 5 |
|---|---|---|---|
| Bar capacity: Round | 2 | 3 | 5 |
| Hexagon | 1-3/4 | 2-9/16 | 4-5/16 |
| Square | 1-7/16 | 2-1/8 | 3-1/2 |
| Swing capacity: Over bedways | 16 | 21 | 21 |
| Over carriage wings | 14 | 19-1/2 | 19-1/2 |
| Over cross slide | 7-5/8 | 11 | 11 |
| Distance from spindle nose to face of hexagon turret | 25 | 36 | 36 |
| Spindle size, ANSI Type A1 or A2 | 8-A1 | 8-A2 | 11-A2 |
| Spindle speeds, number of | 12 | 12 | 12 |
| Spindle speed range, RPM | 30 to 1600 | 30 to 1300 | 75 to 950 |
| Horsepower, main drive, two speed | 18/9 | 15/30 | 15/30 |
| Optional H.P. | --- | 15/30 | 15/30 |
| Longitudinal travel, carriage | 18-1/2 | 26-1/2 | 26-1/2 |
| Cross-slide travel | 9 | 13 | 13 |
| Hexagon turret ram travel | 10 | 15 | 15 |
| Number of feeds | 12 | 12 | 12 |
| Feed range, longitudinal feed, IPR | 0.0025 to 0.050 | 0.0025 to 0.050 | .002 to 0.050 |
| cross feed, IPR | 0.0015 to 0.032 | 0.001 to 0.028 | .001 to 0.028 |
| Diameter of tool holes, hexagon turret | 1-1/2 | 2 | 2 |
| Optional H.P. and spindle speeds (RPM): | 39 to 2000 | 37 to 1600 | 31 to 1170 |

NOTE: Machine sizes and capacities shall not be less than the stated requirements for the size ordered nor greater than the requirements of the next larger size as shown in table I. When the largest size machine is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.





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TABLE II. Alignment and performance tolerances.

| Alignment | Permissible tolerances |
|---|---|
| Bed level in longitudinal direction and transverse direction, convex only | 0.001 inch in 24 inches |
| Axial runout of flange mounting surface of spindle | 0.0005 inch TIR |
| Runout of outside taper pilot diameter of spindle | 0.0005 inch TIR |
| Runout of collet chuck at free end of test mandrel - 2" from collet<br>- 6" from collet | 0.003 inch TIR<br>0.0045 inch TIR |
| Hexagon turret tool holes aligned with spindle axis | 0.0015 |
| Hexagon turret tool hole counterbores aligned with spindle axis | 0.0015 inch TIR |
| Hexagon turret holes parallel with ram slide, with 6-inch mandrel, in vertical plane and horizontal plane | 0.0015 inch in 6 inches |
| Hexagon turret faces square with spindle axis, check points at 180° in vertical plane and horizontal plane | 0.001 inch TIR |
| Automatic stoppage of longitudinal movement by feed trip at the same point within | 0.005 inch |
| Manual feed of ram against stop within | 0.002 inch |
| Repeatability of hexagon turret and index mechanism with test mandrel extended 6 inches from turret face, in vertical plane and horizontal plane | 0.00015 inch |
| Spindle axis parallel with ram slide, with 6-inch test mandrel, in vertical plane and horizontal plane | 0.001 inch in 6 inches (Vertical, high at free end of mandrel) |



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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Navy - SH
DLA - GS
Air Force - 84

User activities:
Army - SM
Navy - MC, YD, AS

Preparing activity:
DLA - IP

Project (3416-0220)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-L-4533F | Lathes, Turret, Horizontal, Universal, Ram Type |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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